

The Latency Race: Registers vs. RAM

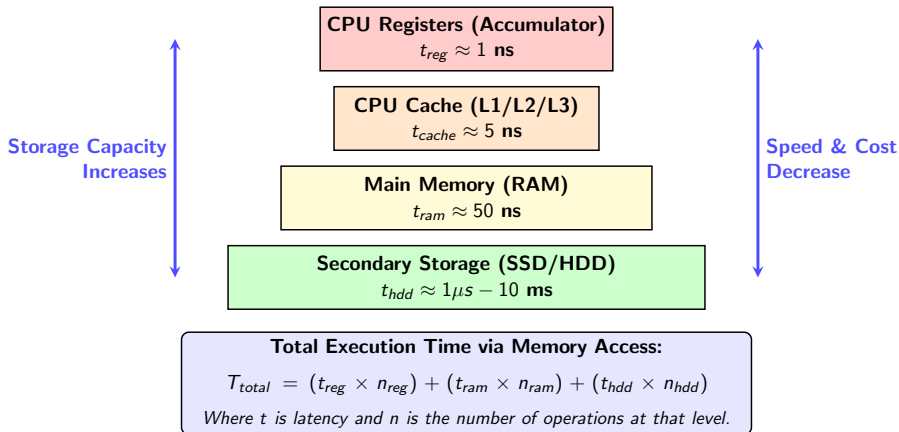
Quiz Review & Memory Hierarchy

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The Memory Hierarchy & I/O Overhead



Quiz Review: Question 1

In computer architecture, which storage location is the fastest for the CPU to access?

- A) The external hard drive
- B) The RAM (Mailboxes)
- C) A CPU Register (like the Accumulator)
- D) Cloud storage

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Answer: C

Registers are built directly into the CPU, meaning the data doesn't have to travel across the motherboard bus.

Quiz Review: Question 2

In the context of the Little Man Computer, what happens during a "STO" instruction?

- A) Data is moved from the slow RAM into the fast Accumulator.
- B) Data is moved from the fast Accumulator across the bus to the slower RAM mailboxes.
- C) The program stops executing completely.
- D) The computer adds a number to the Accumulator.

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Answer: B

"Store" pushes data out of the CPU and into main memory, incurring a latency penalty.

Quiz Review: Question 3

Why did Program A (The RAM Hog) take significantly more instruction cycles to complete the same task as Program B?

- A) Because it had to perform complex multiplication.
- B) Because it used unnecessary STO and LDA instructions to move data in and out of RAM.
- C) Because the user typed the inputs in too slowly.
- D) Because Program A had more INP instructions.

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Answer: B

Every trip to RAM costs time. Program A wasted cycles moving data it was about to use immediately anyway.

Quiz Review: Question 4

What architectural problem does Program A demonstrate by constantly moving data back and forth?

- A) The von Neumann Bottleneck
- B) The Turing Paradox
- C) The Silicon Limit
- D) The Accumulator Overflow

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- B) The Turing Paradox
- C) The Silicon Limit
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Answer: A

The Von Neumann Bottleneck refers to the limited throughput (data transfer rate) between the CPU and memory.

Quiz Review: Question 5

Why was Program B able to completely avoid using the "STO" (Store) command?

- A) Because the program didn't actually do any math.
- B) Because the data was processed and outputted immediately while it was still sitting in the Accumulator.
- C) Because the "OUT" command automatically stores data in RAM.
- D) Because the RAM was full.

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Answer: B

If you don't need to save a variable for later, process it in the register immediately!

Quiz Review: Question 6

If fetching data from RAM takes 50ns, and accessing the Accumulator takes 1ns, what is the best way to reduce latency?

- A) Use the STO command as often as possible.
- B) Minimize STO and LDA commands by doing math on data while it is still in the Accumulator.
- C) Add more INP commands to slow the user down.
- D) Use the HLT command earlier in the program.

Quiz Review: Question 6

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Answer: B

Minimizing n_{ram} in our equation $T_{total} = (t_{reg} \times n_{reg}) + (t_{ram} \times n_{ram})$ drastically lowers total time.

Quiz Review: Question 7

What happens to the existing data in the Accumulator the moment a new "INP" command is executed?

- A) It is automatically saved to mailbox 99.
- B) It is pushed down into a secondary register.
- C) It is completely overwritten and erased by the new input.
- D) It is added to the new input automatically.

Quiz Review: Question 7

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- A) It is automatically saved to mailbox 99.
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- C) It is completely overwritten and erased by the new input.
- D) It is added to the new input automatically.

Answer: C

The Accumulator holds exactly one value. New inputs destroy old data unless you intentionally Store it first.

Quiz Review: Question 8

How many memory access instructions (STO and LDA) did Program A execute compared to Program B?

- A) Program A executed 6, while Program B executed 0.
- B) Both programs executed the exact same number.
- C) Program A executed 0, while Program B executed 6.
- D) Program A executed 3, while Program B executed 3.

Quiz Review: Question 8

How many memory access instructions (STO and LDA) did Program A execute compared to Program B?

- A) Program A executed 6, while Program B executed 0.
- B) Both programs executed the exact same number.
- C) Program A executed 0, while Program B executed 6.
- D) Program A executed 3, while Program B executed 3.

Answer: A

Program A had 3 STO and 3 LDA commands (6 total). Program B handled everything strictly in the CPU.

Quiz Review: Question 9

Optimizing code to behave like Program B helps the CPU utilize what hardware component efficiently?

- A) The cooling fan
- B) The CPU Cache and internal registers
- C) The solid-state drive (SSD)
- D) The graphics card (GPU)

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- C) The solid-state drive (SSD)
- D) The graphics card (GPU)

Answer: B

Modern compilers do exactly this: they try to keep variables inside fast CPU Cache and Registers to avoid hitting the slower RAM.

Quiz Review: Question 10

If a program requires you to use a user's inputted number multiple times throughout a long program, what must you do despite the latency cost?

- A) Output it immediately.
- B) Use the STO command to save it to RAM so it isn't overwritten.
- C) Ask the user to input the number again every time.
- D) Use the HLT command.

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Answer: B

Latency is the cost of doing business when we need persistent variables. We accept the t_{ram} delay to ensure data isn't lost!

Great Job! Any questions on Memory Hierarchy?